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VAPT Assignment -1

Whois Enumeration

```
4s ~  
• ▶ whois hackthebox.com
Domain Name: HACKTHEBOX.COM
Registry Domain ID: 1589213536_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.registrar.amazon
Registrar URL: http://registrar.amazon.com
Updated Date: 2026-05-15T15:03:57Z
Creation Date: 2010-03-18T03:26:29Z
Registry Expiry Date: 2027-03-18T03:26:29Z
Registrar: Amazon Registrar, Inc.
Registrar IANA ID: 468
Registrar Abuse Contact Email: trustandsafety@support.aws.com
Registrar Abuse Contact Phone: +1.2024422253
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Name Server: CODY.NS.CLOUDFLARE.COM
Name Server: JILL.NS.CLOUDFLARE.COM
DNSSEC: signedDelegation
DNSSEC DS Data: 2371 13 2 94504539EEB9E894BBBD9F27F1A0B6E0BE43E36106EBC14329A34544CD9EC56B
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-06-22T17:10:15Z <<<

For more information on Whois status codes, please visit https://icann.org/epp

NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiration
date of the domain name registrant's agreement with the sponsoring
registrar. Users may consult the sponsoring registrar's Whois database to
view the registrar's reported date of expiration for this registration.

TERMS OF USE: You are not authorized to access or query our Whois
database through the use of electronic processes that are high-volume and
automated except as reasonably necessary to register domain names or
modify existing registrations; the Data in VeriSign Global Registry
Services' ("VeriSign") Whois database is provided by VeriSign for
information purposes only, and to assist persons in obtaining information
about or related to a domain name registration record. VeriSign does not
```

Whois is used to get the information of a domain, it's ip, registrar, registry etc

DNS Enumeration using dig

```
0s ~ dig hackthebox.com

; <<>> DiG 9.20.24 <<>> hackthebox.com
;; global options: +cmd
;; Got answer:
;; -->HEADER<-- opcode: QUERY, status: NOERROR, id: 50328
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1280
;; QUESTION SECTION:
;hackthebox.com.                IN      A

;; ANSWER SECTION:
hackthebox.com.                300     IN      A      109.176.239.70
hackthebox.com.                300     IN      A      109.176.239.69

;; Query time: 81 msec
;; SERVER: 10.168.109.153#53(10.168.109.153) (UDP)
;; WHEN: Mon Jun 22 22:42:39 IST 2026
;; MSG SIZE rcvd: 75
```

Dig is used to query the DNS Servers and debug them

Mail Server Enumeration

```
0s ~ dig mx amrita.edu

; <<>> DiG 9.20.24 <<>> mx amrita.edu
;; global options: +cmd
;; Got answer:
;; -->HEADER<-- opcode: QUERY, status: NOERROR, id: 52805
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

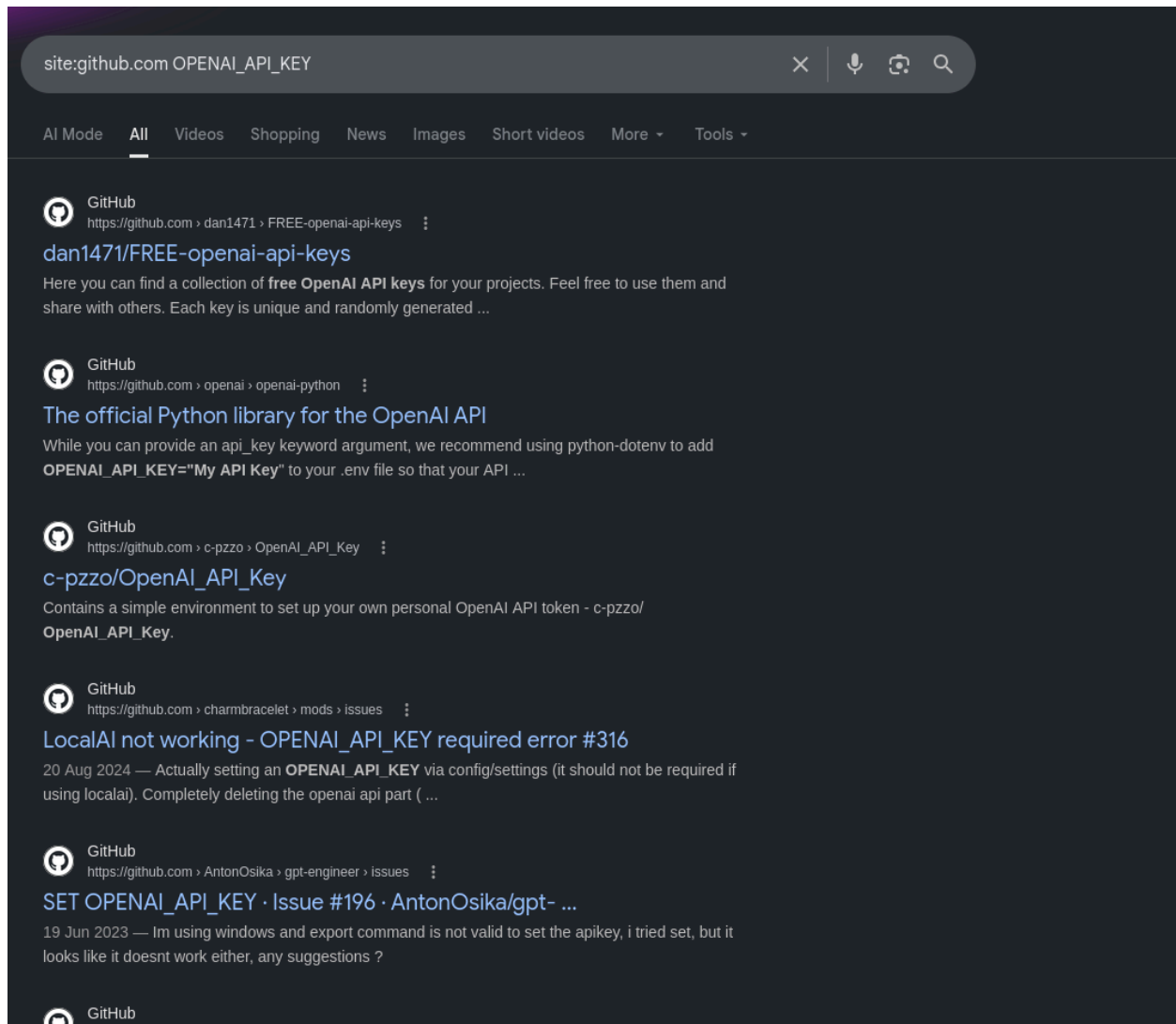
;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1280
;; QUESTION SECTION:
;amrita.edu.                    IN      MX

;; ANSWER SECTION:
amrita.edu.                    3600    IN      MX      0 amrita-edu.mail.protection.outlook.com.

;; Query time: 167 msec
;; SERVER: 10.168.109.153#53(10.168.109.153) (UDP)
;; WHEN: Mon Jun 22 22:43:40 IST 2026
;; MSG SIZE rcvd: 93
```

Dig mx is used to query the mail exchange records of a domain

Google Dorking and search engine reconnaissance



Filetype .md

site:github.com filetype:md



AI Mode

All

Shopping

Images

Short videos

Forums

Videos

More ▾

Tools ▾



GitHub

[https://github.com > AzurePublicDataset > blob > master](https://github.com/AzurePublicDataset/blob/master) ⋮

AzurePublicDataset/AzureFunctionsDataset2019.md at ...

Functions in Azure Functions are grouped into Applications. Included here is only data pertaining to a random sample of Azure Functions applications. The ...



GitHub

[https://github.com > bytebase > dbhub > blob > main > C...](https://github.com/bytebase/dbhub/blob/main/C...) ⋮

CLAUDE.md - bytebase/dbhub

Build: pnpm run build - Compiles TypeScript to JavaScript using tsup; Start: pnpm run start - Runs the compiled server; Dev: pnpm run dev - Runs server with ...



GitHub

[https://github.com > README_CN](https://github.com/README_CN) · [Translate this page](#) ⋮

电力变压器数据集(ETDataset)

电力变压器数据集(ETDataset) · ETT-small : 含有2个电力变压器 (来自2个站点) 的数据, 包括负载、油温。 · ETT-large : 含有39个电力变压器 (来自39个站点) 的数据, 包括负载、油温 ...



GitHub

[https://github.com > polostbiop > oup > blob > index106...](https://github.com/polostbiop/oup/blob/index106...) ⋮

oup/index10654.md at main · polostbiop/oup

Contribute to **polostbiop/oup** development by creating an account on GitHub.



GitHub

[https://github.com > NREL > buildstockbatch > blob > R...](https://github.com/NREL/buildstockbatch/blob/R...) ⋮

buildstockbatch/README.md at develop

This set of tools provides a means to **run and manage batch simulations** for the ResStock and ComStock building stock energy models.



GitHub

[https://github.com > blob > master](https://github.com/blob/master) · [Translate this page](#) ⋮

Netcraft Intelligence Gathering

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Background

Site title	Reddit - Please wait for verification	Date first seen	June 2002
Site rank	2157	Primary language	English
Description	Not Present		

Network

Site	http://reddit.com	Domain	reddit.com
Netblock Owner	Fastly, Inc.	Nameserver	ns-557.awsdns-05.net
Hosting company	Fastly	Domain registrar	markmonitor.com
Hosting country	US	Nameserver organisation	whois.markmonitor.com
IPv4 address	151.101.129.140 (VirusTotal)	Organisation	Reddit Inc., United States
IPv4 autonomous systems	AS54113	DNS admin	awsdns-hostmaster@amazon.com
IPv6 address	2a04:4e42:400:0:0:0:396	Top Level Domain	Commercial entities (.com)
IPv6 autonomous systems	AS54113	DNS Security Extensions	Enabled
Reverse DNS	Unknown		

IP delegation

IPv4 address (151.101.129.140)

IP range	Country	Name	Description
::ffff:0:0:0:0/96	United States	IANA-IPV4-MAPPED-ADDRESS	Internet Assigned Numbers Authority
↳ 151.101.0.0-151.101.255.255	United States	SKYCA-3	Fastly, Inc.
↳ 151.101.129.140	United States	SKYCA-3	Fastly, Inc.

IPv6 address (2a04:4e42:400:0:0:0:396)

IP range	Country	Name	Description
::/0	N/A	ROOT	Root inet6num object
↳ 2a00::/11	European Union	EU-ZZ-2A00	RIPE NCC
↳ 2a00::/12	Netherlands	EU-ZZ-2A00	RIPE Network Coordination Centre
↳ 2a04:4e40::/29	European Union	US-FASTLY-20130718	Fastly, Inc.
↳ 2a04:4e42:400:0:0:0:396	European Union	US-FASTLY-20130718	Fastly, Inc.

SSL/TLS

This is not a HTTPS site. If you're looking for SSL/TLS information try the [HTTPS site report](#).

Netcraft website is used to gather all the data of a domain like hosting services, public IPS, SPF, DMARC, technologies used, etc

Shodan Search engine

The screenshot displays the Shodan search engine interface. At the top, there's a navigation bar with links for SHODAN, Explore, Downloads, Pricing, and an Account button. The search bar contains the query 'country:IN port:80 Response:200'. Below the search bar, the interface is divided into several sections:

- TOTAL RESULTS:** 356
- TOP CITIES:** A list of cities with their respective result counts: Mumbai (247), Hyderabad (27), Doddaballapura (18), Chennai (12), and New Delhi (10).
- TOP ORGANIZATIONS:** A list of organizations with their respective result counts: Amazon Data Services India (131), Amazon.com, Inc. (74), Akamai Technologies, Inc. (47), Amazon Technologies Inc. (33), and DigitalOcean, LLC (18).
- TOP PRODUCTS:** A list of products with their respective result counts: nginx (108), AWS ELB (25), Apache httpd (17), Jellyfin (8), and Microsoft IIS httpd (1).

The main content area shows detailed search results for the query. It includes a 'Product Spotlight' for 'Passbolt | Open source password manager for teams'. Below this, there are three search results, each with a cloud icon and a 'cloud' button. The first result is for '3.109.132.13' (en2-3-109-132-13.ap-south-1.compu) with a date of Mon, 22 Jun 2026 17:07:15 GMT. The second result is for '40.81.243.219' (Microsoft Corporation) with a date of Mon, 22 Jun 2026 16:27:37 GMT. The third result is for '13.206.242.10' (en2-13-206-242-10.ap-south-1.compu) with a date of Mon, 22 Jun 2026 15:37:13 GMT. Each result includes a 'View Report' button and a 'View on Map' button.

Shodan is a search engine for Internet connected devices, you can even look up webcams

Subdomain Enumeration using The harvester

Go Assetfinder

The Xmas Discord server can be found here: <https://discord.gg/AMryu>

```
0s ~ 
• ▶ assetfinder github.com
github.com
accelerator.github.com
accessibility.github.com
accessibility-playbook.github.com
action.github.com
admin.github.com
alive.github.com
alive-staging.github.com
api.github.com
archiveprogram.github.com
asset-generator.github.com
atom-installer.github.com
awesome-copilot.github.com
blog.github.com
blog-freeze.github.com
bounty.github.com
boxen.github.com
brand.github.com
bug-bash.github.com
campus.github.com
cdn-185-199-108-133.github.com
cdn-185-199-108-153.github.com
cdn-185-199-108-154.github.com
cdn-185-199-109-133.github.com
cdn-185-199-109-153.github.com
cdn-185-199-109-154.github.com
cdn-185-199-110-133.github.com
cdn-185-199-110-153.github.com
cdn-185-199-111-133.github.com
cdn-185-199-111-153.github.com
cdn-185-199-111-154.github.com
central.github.com
classroom.github.com
cli.github.com
f.cloud.github.com
codeload.github.com
```

Host Discovery

Nmap

```
10s ~ 
x ▶ sudo nmap -sN 10.129.3.9
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-22 23:04 +0530
Nmap scan report for 10.129.3.9
Host is up (0.076s latency).
Not shown: 998 closed tcp ports (reset)
PORT      STATE      SERVICE
22/tcp    open|filtered ssh
80/tcp    open|filtered http

Nmap done: 1 IP address (1 host up) scanned in 15.18 seconds
15s ~ 
• ▶ |
```

I'm using an IP of a hackthebox machine for enumeration

```
0s ~ 
x ▶ nmap 10.129.3.9
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-22 23:01 +0530
Nmap scan report for 10.129.3.9
Host is up (0.077s latency).
Not shown: 998 closed tcp ports (conn-refused)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http

Nmap done: 1 IP address (1 host up) scanned in 1.70 seconds
1s ~ 
• ▶ |
```

Service Enumeration and version detection

```

3s ~ 
x ▶ nmap -sV 10.129.3.9
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-22 23:03 +0530
Nmap scan report for 10.129.3.9
Host is up (0.074s latency).
Not shown: 998 closed tcp ports (conn-refused)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 9.6p1 Ubuntu 3ubuntu13.16 (Ubuntu Linux; protocol 2.0)
80/tcp    open  http     nginx 1.24.0 (Ubuntu)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 8.80 seconds
8s ~ 
• ▶ 

```

Banner Grabbing

```

8s ~ 
• ▶ sudo nmap -A 10.129.3.9
[sudo] password for praneesh:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-22 23:05 +0530
Nmap scan report for 10.129.3.9
Host is up (0.14s latency).
Not shown: 998 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 9.6p1 Ubuntu 3ubuntu13.16 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
| 256 eb:ab:8f:be:99:02:0b:3e:c4:1c:83:b2:66:2f:17:13 (ECDSA)
|_ 256 c1:69:ab:84:f3:88:8b:b3:8a:ae:e2:28:35:54:35:0b (ED25519)
80/tcp    open  http     nginx 1.24.0 (Ubuntu)
|_http-title: Did not follow redirect to http://nimbus.htb/
|_http-server-header: nginx/1.24.0 (Ubuntu)
Device type: general purpose
Running: Linux 4.X|5.X
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:5
OS details: Linux 4.15 - 5.19
Network Distance: 2 hops
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

TRACEROUTE (using port 8888/tcp)
HOP RTT      ADDRESS
1 117.53 ms 10.10.14.1
2 117.64 ms 10.129.3.9

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 16.82 seconds
19s ~ 
• ▶ 

```

Webtechnology fingerprinting

```
15s ~ 
• ▶ whatweb https://praneeshrv.me
https://praneeshrv.me [307 Temporary Redirect] HTTPServer[Vercel], IP[64:ff9b::d8c6:4f01], RedirectLocation[https://www.praneeshrv.me/], Strict-Transport-Security[max-age=63072000], UncommonHeaders[x-vercel-id]
https://www.praneeshrv.me/ [200 OK] Email[praneeshrv404@gmail.com,you@example.com], HTML5, HTTPServer[Vercel], IP[64:ff9b::d8c6:4f01], Meta-Author[Praneesh R V], Open-Graph-Protocol[website], Script[application/ld+json], Strict-Transport-Security[max-age=63072000], Title[Archive of the Shattered Star - Praneesh R V], UncommonHeaders[access-control-allow-origin,x-matched-path,x-nextjs-prerender,x-nextjs-stale-time,x-vercel-cache,x-vercel-id]
15s ~ 
• ▶ 
```

HTTP- title Enumeration

```
• ▶ nmap --script http-title www.google.com
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-22 23:15 +0530
Nmap scan report for www.google.com (142.251.155.119)
Host is up (0.12s latency).
Other addresses for www.google.com (not scanned): 2001:4860:482c:7700:: 2001:4860:4827:7700:: 2001:4860:482b:7700:: 2001:4860:4829:7700:: 2001:4860:482d:7700:: 2001:4860:4826:7700:: 2001:4860:4828:7700:: 2001:4860:482a:7700:: 142.251.154.119 142.251.156.119 142.251.153.119 142.251.152.119 142.251.157.119 142.251.150.119 142.251.151.119
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http
|_http-title: Google
443/tcp   open  https
```

Http-resource enumeration

```
• ▶ nmap -Pn -T4 --min-rate 1000 --script http-enum scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-22 23:23 +0530
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.38s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 992 closed tcp ports (conn-refused)
PORT      STATE SERVICE
22/tcp    open  ssh
25/tcp    filtered smtp
80/tcp    open  http
| http-enum:
|_ /images/: Potentially interesting directory w/ listing on 'apache/2.4.7 (ubuntu)'
135/tcp    filtered msrpc
139/tcp    filtered netbios-ssn
445/tcp    filtered microsoft-ds
9929/tcp   open  nping-echo
31337/tcp  open  Elite
Nmap done: 1 IP address (1 host up) scanned in 38.69 seconds
38s ~ 
• ▶ 
```

SMB enumeration

```
x ▶ enum4linux -a 10.129.9.107
Starting enum4linux v0.9.1 ( http://labs.portcullis.co.uk/application/enum4linux/ ) on Mon Jun 2
2 23:30:11 2026

=====( Target Information )=====
=====
Target ..... 10.129.9.107
RID Range ..... 500-550,1000-1050
Username ..... ''
Password ..... ''
Known Usernames .. administrator, guest, krbtgt, domain admins, root, bin, none

=====( Enumerating Workgroup/Domain on 10.129.9.107 )=====
=====
Can't load /etc/samba/smb.conf - run testparm to debug it

[E] Can't find workgroup/domain

=====( Nbtstat Information for 10.129.9.107 )=====
=====
Can't load /etc/samba/smb.conf - run testparm to debug it
Looking up status of 10.129.9.107
No reply from 10.129.9.107

=====( Session Check on 10.129.9.107 )=====
=====

[E] Server doesn't allow session using username '', password ''. Aborting remainder of tests.
```

SNMP Discovery

```
0s ~ 
x ▶ sudo nmap -sU -p 161 scanme.nmap.org
[sudo] password for praneesh:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-22 23:32 +0530
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.050s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f

PORT      STATE      SERVICE
161/udp   open|filtered snmp

Nmap done: 1 IP address (1 host up) scanned in 1.85 seconds
3s ~ 
```

Security Headers

[Home](#) [About](#) [API](#)

 **Security Headers**
by snyk

Scan your site now

Scan

☐ Hide results ☒ Follow redirects

Security Report Summary



Site:

<http://scanme.nmap.org/> - [\(Scan again over https\)](#)

IP Address:

45.33.32.156

Report Time:

22 Jun 2026 17:59:17 UTC

Headers:

✖ Content-Security-Policy

✖ X-Frame-Options

✖ X-Content-Type-Options

✖ Referrer-Policy

✖ Permissions-Policy

Warning:

Grade capped at A, please see warnings below.

Advanced:

Ouch, you should work on your security posture immediately:

Start Now

Missing Headers	
Content-Security-Policy	Content Security Policy is an effective measure to protect your site from XSS attacks. By whitelisting sources of approved content, you can prevent the browser from loading malicious assets.
X-Frame-Options	X-Frame-Options tells the browser whether you want to allow your site to be framed or not. By preventing a browser from framing your site you can defend against attacks like clickjacking. Recommended value "X-Frame-Options: SAMEORIGIN".
X-Content-Type-Options	X-Content-Type-Options stops a browser from trying to MIME-sniff the content type and forces it to stick with the declared content-type. The only valid value for this header is "X-Content-Type-Options: nosniff".
Referrer-Policy	Referrer Policy is a new header that allows a site to control how much information the browser includes with navigations away from a document and should be set by all sites.
Permissions-Policy	Permissions Policy is a new header that allows a site to control which features and APIs can be used in the browser.

LLM Aided Enumeration

analyze whois information of github.com and summarize important reconnaissance findings

WHOIS Analysis of GitHub ↗

Scope: Passive reconnaissance using publicly available WHOIS registration data only. No active scanning or intrusive techniques.

Executive Summary

The WHOIS record for GitHub ↗ reveals a mature, well-protected corporate domain owned by GitHub, Inc. ↗. The registration uses enterprise-grade domain management through MarkMonitor and employs multiple defensive controls to reduce the risk of domain hijacking. The DNS infrastructure is distributed across both NSONE and AWS Route 53, indicating a resilient and highly available architecture. 🔥 Whois +1

Key WHOIS Findings

Finding	Value	Recon Significance
Domain	github.com	Primary GitHub web property
Organization	GitHub, Inc.	Confirms ownership
Registrar	MarkMonitor Inc.	Common among Fortune 500 companies
Creation Date	2007-10-09	Long-established domain
Expiry Date	2026-10-09	Actively maintained

Domain	github.com	Primary GitHub web property
Organization	GitHub, Inc.	Confirms ownership
Registrar	MarkMonitor Inc.	Common among Fortune 500 companies
Creation Date	2007-10-09	Long-established domain
Expiry Date	2026-10-09	Actively maintained
Last Updated	2024-09-07	Recent administrative activity
Registrant Country	United States	Geographic attribution
Nameservers	NSONE + AWS Route 53	Redundant DNS architecture
Domain Locks	Delete, Transfer, Update Prohibited	Strong anti-hijacking protection

Infrastructure Intelligence

1. Enterprise Registrar

GitHub uses MarkMonitor as its registrar. MarkMonitor is frequently used by large technology companies because it specializes in:

- Brand protection
- Domain portfolio management
- Anti-phishing operations
- High-security registrar services

Recon implication: The domain is unlikely to be vulnerable to simple registrar-targeted attacks.  Whois +1

2. DNS Architecture

WHOIS exposes eight authoritative nameservers:

NSONE

- dns1.p08.nsone.net
- dns2.p08.nsone.net
- dns3.p08.nsone.net
- dns4.p08.nsone.net

AWS Route 53

- ns-1283.awsdns-32.org
- ns-1707.awsdns-21.co.uk
- ns-421.awsdns-52.com
- ns-520.awsdns-01.net

Recon implication:

- Multi-provider DNS increases resilience.
- Reduces single points of failure.
- Suggests a globally distributed DNS strategy.
- Indicates likely use of cloud-native infrastructure and traffic engineering.  Whois +1

3. Domain Security Controls

The WHOIS record shows:

- clientDeleteProhibited
- clientTransferProhibited
- clientUpdateProhibited



These are registrar-enforced locks.

Potential Follow-on Reconnaissance Opportunities

Based on WHOIS alone, a security analyst might next investigate:

1. DNS records (A, AAAA, MX, TXT, CNAME).
2. Subdomain enumeration.
3. Certificate Transparency logs.
4. ASN and IP ownership.
5. Cloud service usage patterns.
6. Historical DNS changes.
7. Related GitHub-owned domains such as [github.io](#), [githubusercontent.com](#), and [githubassets.com](#).

These would provide additional infrastructure visibility without requiring intrusive activity.

Security Assessment

Overall maturity: High

Indicators:

- ✓ Enterprise registrar (MarkMonitor)
- ✓ Registrar lock protections enabled
- ✓ Distributed DNS providers (NSONE + AWS)
- ✓ Long-lived domain ownership
- ✓ Active maintenance and updates
- ✓ Limited public contact exposure

No obvious weaknesses are visible from the WHOIS record itself. The data reflects a professionally managed corporate domain with standard enterprise security controls in place.  Whois +2

   Sources